Solution Requirements

Date	15 March 2026
Team ID	xxxxxx
Project Name	Comprehensive Measure of well-being
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	The system allows users to securely access the HDI Predictor application.	Medium
2	Authorization levels	All users can access the prediction, analysis, and about pages. Future versions can support Admin and User roles.	Medium
3	External interfaces	The system reads the HDI dataset (CSV), loads the trained Pickle model, and displays graphs generated using Matplotlib and Seaborn.	High

4	Transactions processing	The application accepts Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, and GNI per Capita from the user, validates the inputs, processes them through the machine learning model, and returns the predicted HDI score and category.	High
5	Reporting	The system displays the predicted HDI score, HDI category, and data analysis graphs including Heatmap, Distribution Plot, Scatter Plot, Pair Plot, and Strip Plot.	High
6	Business rules, etc.	All input values must be numeric and within valid ranges. The model predicts HDI using Linear Regression, and the predicted score is classified into Very High, High, Medium, or Low Human Development.	High
7	Compliance to laws or regulations	The application does not collect or store personal user data. It uses only publicly available HDI datasets and follows ethical data usage practices	High
8	*Other*	The application provides an intuitive web interface with Home, Prediction, Analysis, and About pages for easy navigation.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The application should generate predictions quickly after user input.	Prediction response time less than **2 seconds**
2	Scalability	The system should support future expansion with larger datasets or additional machine learning models.	Supports increased dataset size and multiple prediction requests
3	Security & Data Privacy	User inputs are processed securely without storing personal information. Input validation is implemented to avoid invalid data.	Secure data handling with no personal data storage
4	Reliability & Availability	The application should consistently provide accurate predictions and remain available during usage.	**99% application availability** during normal operation
5	Usability & Accessibility	The web interface should be simple, responsive, and easy to navigate for students, researchers, and policymakers.	User-friendly interface compatible with desktop and mobile browsers
6	*Other*	The project should be maintainable and portable, allowing future updates and deployment on different operating systems.	Modular code with cross-platform compatibility